

ENGINEERING MATHEMATICS: COMPREHENSIVE MASTER TEXTBOOK

1. Linear Algebra: Matrices & Systems

Eigenvalues and Eigenvectors

For a square matrix A , the characteristic equation is $\det(A - \lambda I) = 0$. The roots λ are the eigenvalues, and the corresponding vectors are eigenvectors.

$$Av = \lambda v$$

2. Calculus: Multiple Integrals

Double and Triple Integrals

Multiple integrals are used for finding volume and center of mass. Changing order of integration is a standard technique.

3. Differential Equations

First Order Linear Equations

Equations of the form $dy/dx + P(x)y = Q(x)$ are solved using the Integrating Factor (IF) = $e^{\int P dx}$.

$$y * IF = \int (Q * IF) dx + C$$

4. Long-Answer Problem Bank

Problem 1 (Matrices): Find eigenvalues of $A = \begin{bmatrix} 1 & 2 \\ 0 & 2 \end{bmatrix}$.

Step-by-Step Solution:

1. Characteristic equation: $\det(A - \lambda I) = 0$.
2. $|(1-\lambda) \ 2| = 0 \Rightarrow (1-\lambda)(2-\lambda) - 0 = 0$.
3. $\lambda^2 - 3\lambda + 2 = 0 \Rightarrow (\lambda-1)(\lambda-2) = 0$.
4. Eigenvalues: $\lambda_1 = 1, \lambda_2 = 2$.

Problem 2 (Differential Equations): Solve $dy/dx + y/x = x^2$.

Step-by-Step Solution:

1. $P = 1/x$, $Q = x^2$.

2. $IF = e^{\int (1/x) dx} = e^{\ln x} = x$.

3. $y * x = \int (x^2 * x) dx = \int x^3 dx = x^4/4 + C$.

4. $y = x^3/4 + C/x$.

5. Probability and Statistics

Normal Distribution

The probability density function is $f(x) = (1/\sigma\sqrt{2\pi}) * \exp(-(x-\mu)^2 / 2\sigma^2)$.